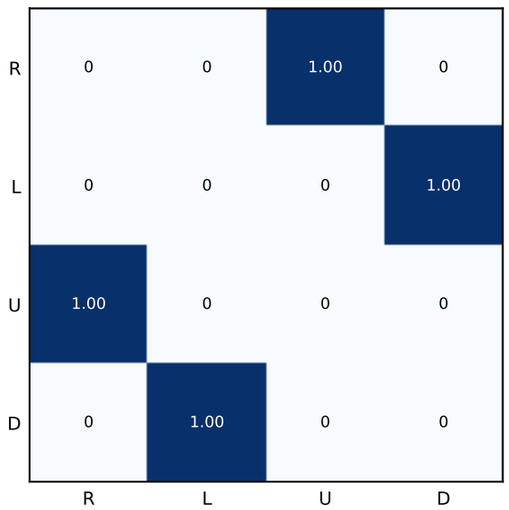
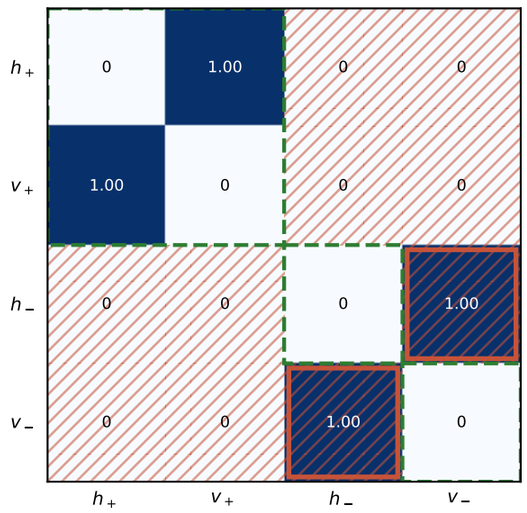
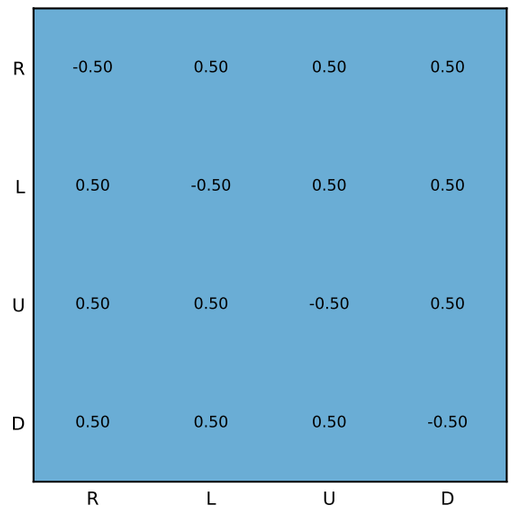
P (permutation)
physical, |C| colour



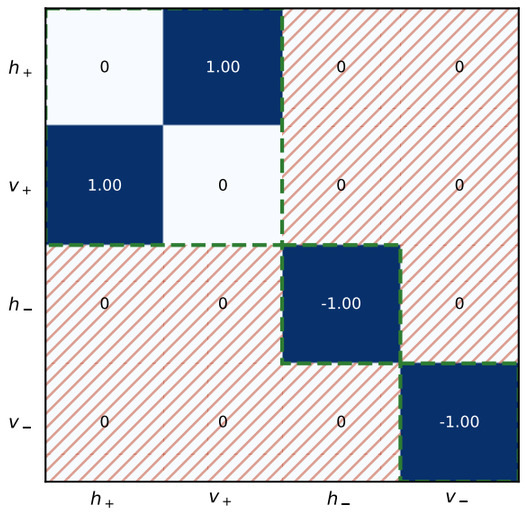
P (permutation)
coherent: $\mathcal{L}_{\mathrm{SW}}$=0.500, $\mathcal{L}^{*}_{\mathrm{SW}}$=0.500



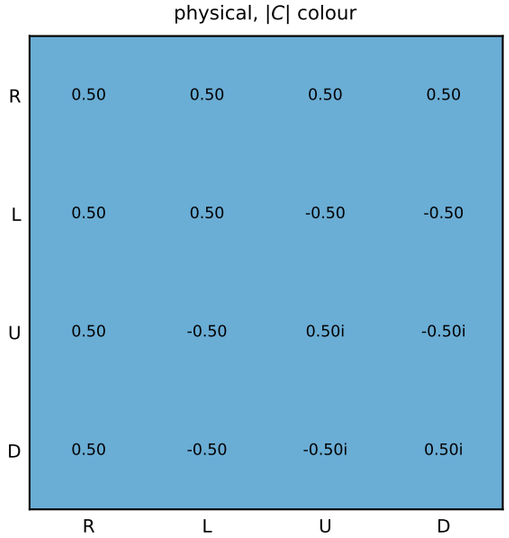
G (Grover)
physical, |C| colour



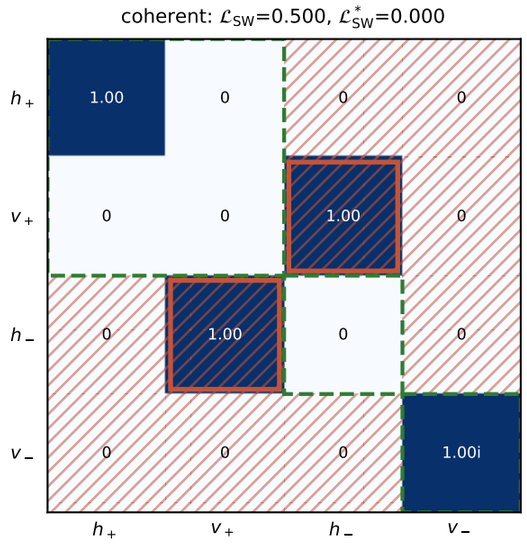
G (Grover)
coherent: $\mathcal{L}_{\mathrm{SW}}$=0.000, $\mathcal{L}^{*}_{\mathrm{SW}}$=0.000



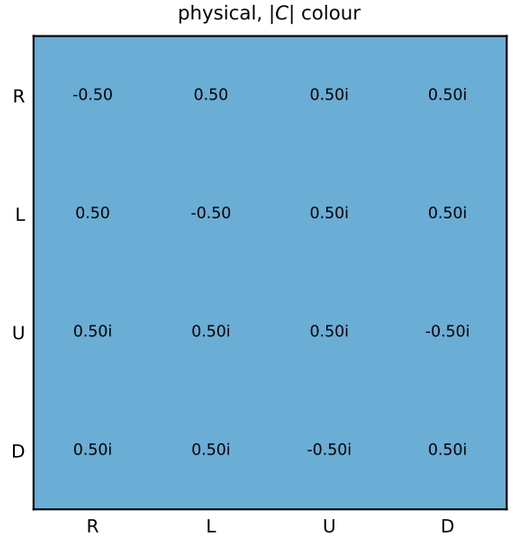
F_c (compass Fourier)
physical, |C| colour



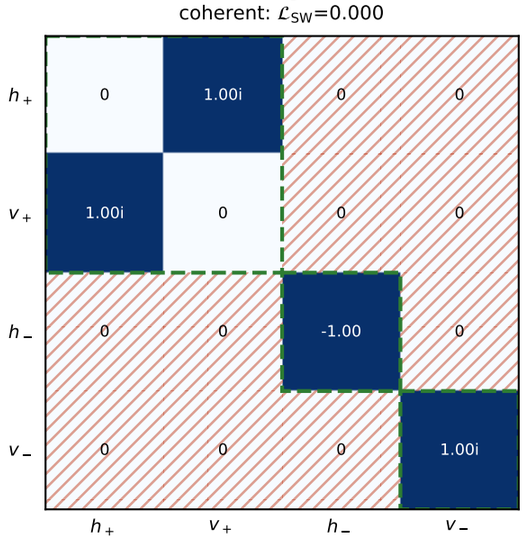
F_c (compass Fourier)
coherent: $\mathcal{L}_{\mathrm{SW}}$=0.500, $\mathcal{L}^{*}_{\mathrm{SW}}$=0.000



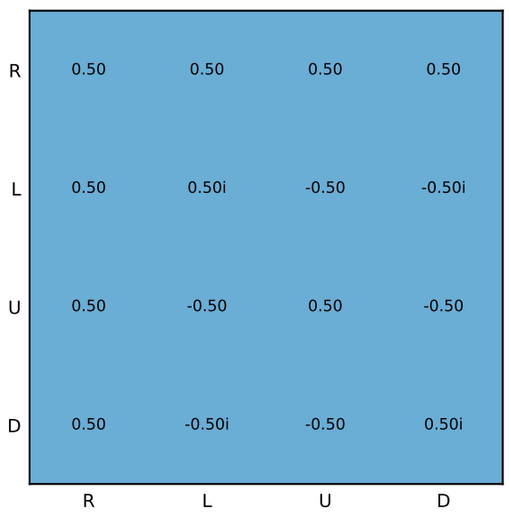
DF_cD, $D = \mathrm{diag}(i, -i, 1, 1)$
physical, |C| colour



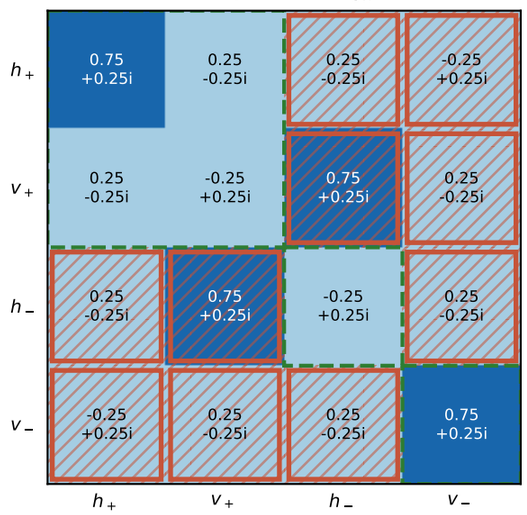
DF_cD, $D = \mathrm{diag}(i, -i, 1, 1)$
coherent: $\mathcal{L}_{\mathrm{SW}}$=0.000



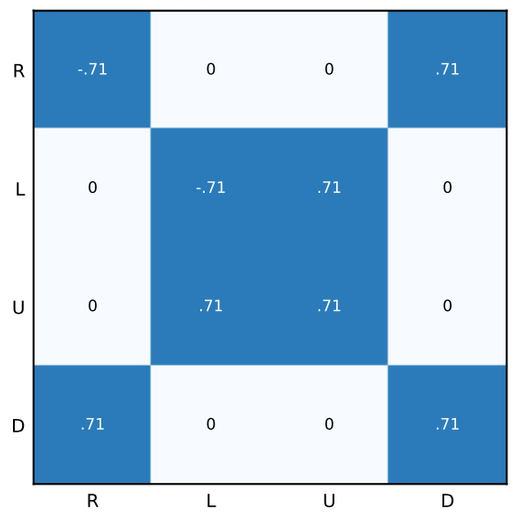
F_p (physical-order Fourier)
physical, |C| colour



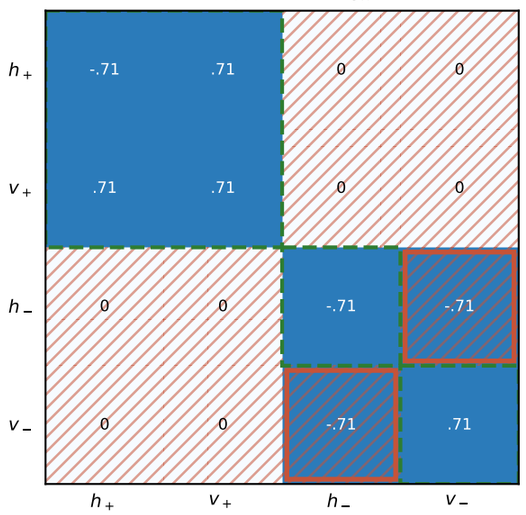
F_p (physical-order Fourier)
coherent: $\mathcal{L}_{\mathrm{SW}}$=0.562, $\mathcal{L}^{*}_{\mathrm{SW}}$=0.073



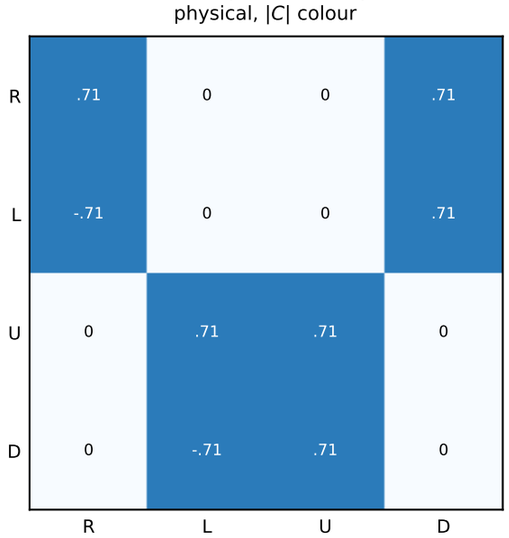
B_{I} (Bell, earlier report)
physical, |C| colour



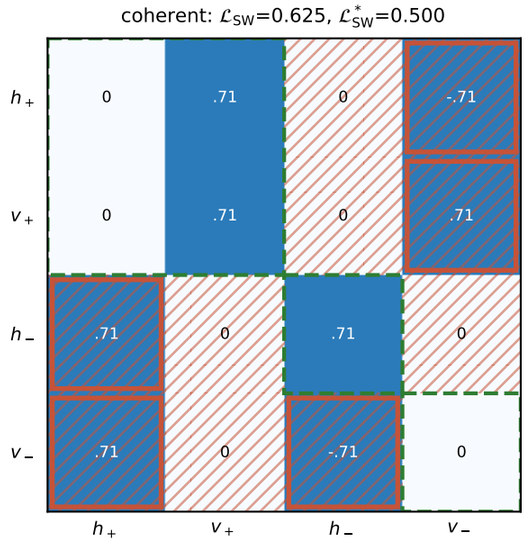
B_{I} (Bell, earlier report)
coherent: $\mathcal{L}_{\mathrm{SW}}$=0.250, $\mathcal{L}^{*}_{\mathrm{SW}}$=0.250



B_{II} (Bell, reproduces IPR plot)
physical, |C| colour



B_{II} (Bell, reproduces IPR plot)
coherent: $\mathcal{L}_{\mathrm{SW}}$=0.625, $\mathcal{L}^{*}_{\mathrm{SW}}$=0.500



How to read each pair of squares

Left: the coin in the direction basis (R,L,U,D).
A column is the output produced by one input direction.

Right: the same coin in the channels $h_\pm = (R \pm L)/\sqrt{2}$,
$v_\pm = (U \pm D)/\sqrt{2}$, ordered (h_+, v_+, h_-, v_-).
Hatched cells change the spatial envelope (forbidden);
a red frame marks a nonzero forbidden amplitude.
Dashed green boxes: the allowed $2+1+1$ blocks.

$\mathcal{L}_{\mathrm{SW}} = \frac{1}{4} \sum_{\text{forbidden}} |\bar{C}_{ij}|^2$ is the mean probability
per step of leaving the fixed standing-wave sectors.

$\mathcal{L}^{*}_{\mathrm{SW}}$: same after the best gauge $C \to DCD$
(numerical minimum for P, F_p, B_{I}, B_{II}).

Zero in a square is sufficient for an exact block solution;
a nonzero value is not a proof of non-integrability
(see P: $\mathcal{L}_{\mathrm{SW}} = \mathcal{L}^{*}_{\mathrm{SW}} = 1/2$, exactly solvable by routing).